



Introduction

- Security is an important concern in homes, offices, and industries. Theft and unauthorized entry can cause serious problems. To overcome this issue, an electronic security system can be used to detect motion and alert the owner immediately.
- This project uses a PIR motion sensor, Arduino Uno, and a GSM SIM800 module. When motion is detected, the system automatically sends an SMS alert and makes a phone call to the owner's mobile phone. This helps the user know about the intrusion instantly.





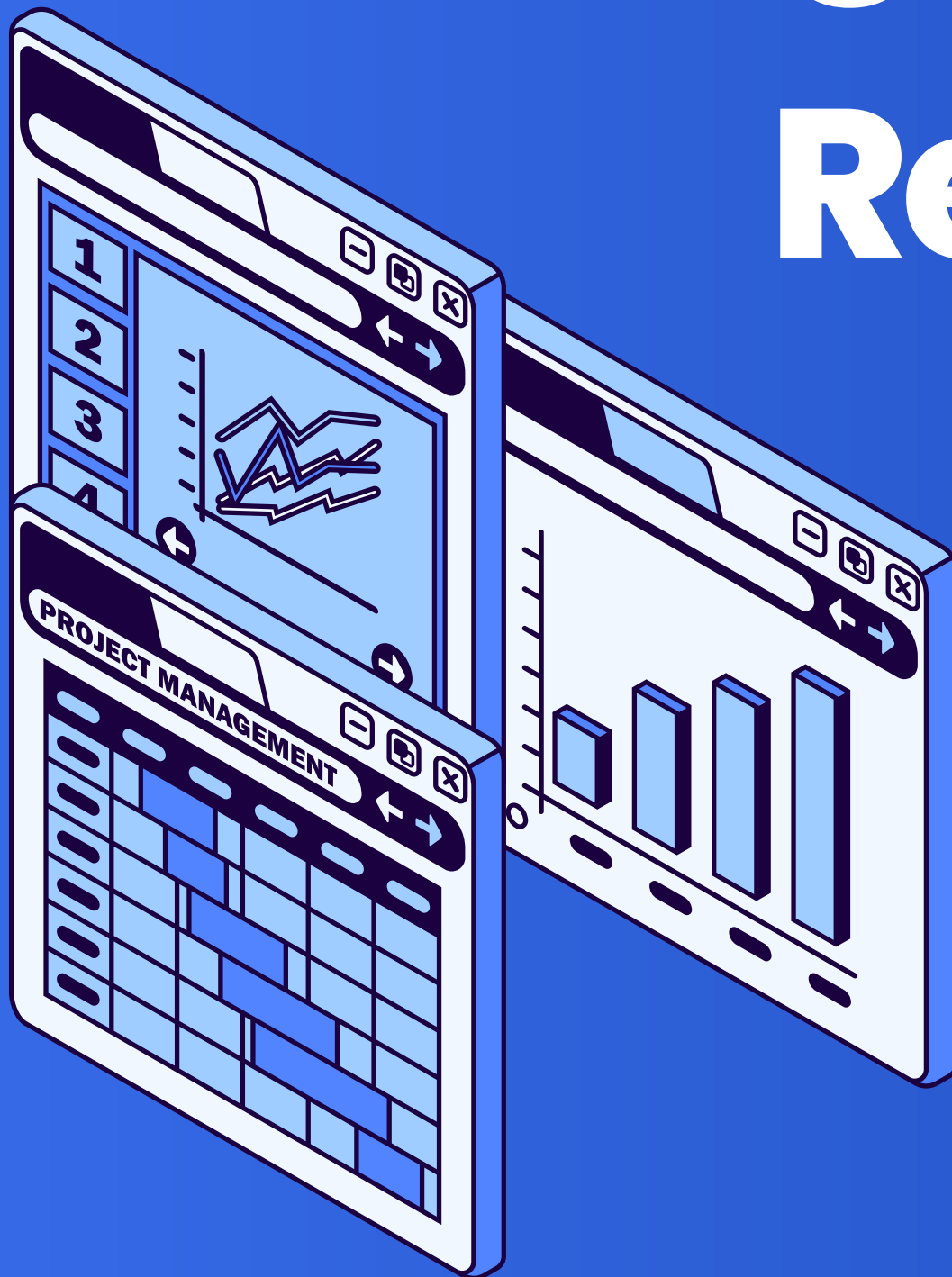
Objective

The main objectives of this project are:

- To detect unauthorized motion using a PIR sensor
- To send an alert SMS to the owner
- To make an automatic phone call during theft detection
- To provide a low-cost and reliable security system



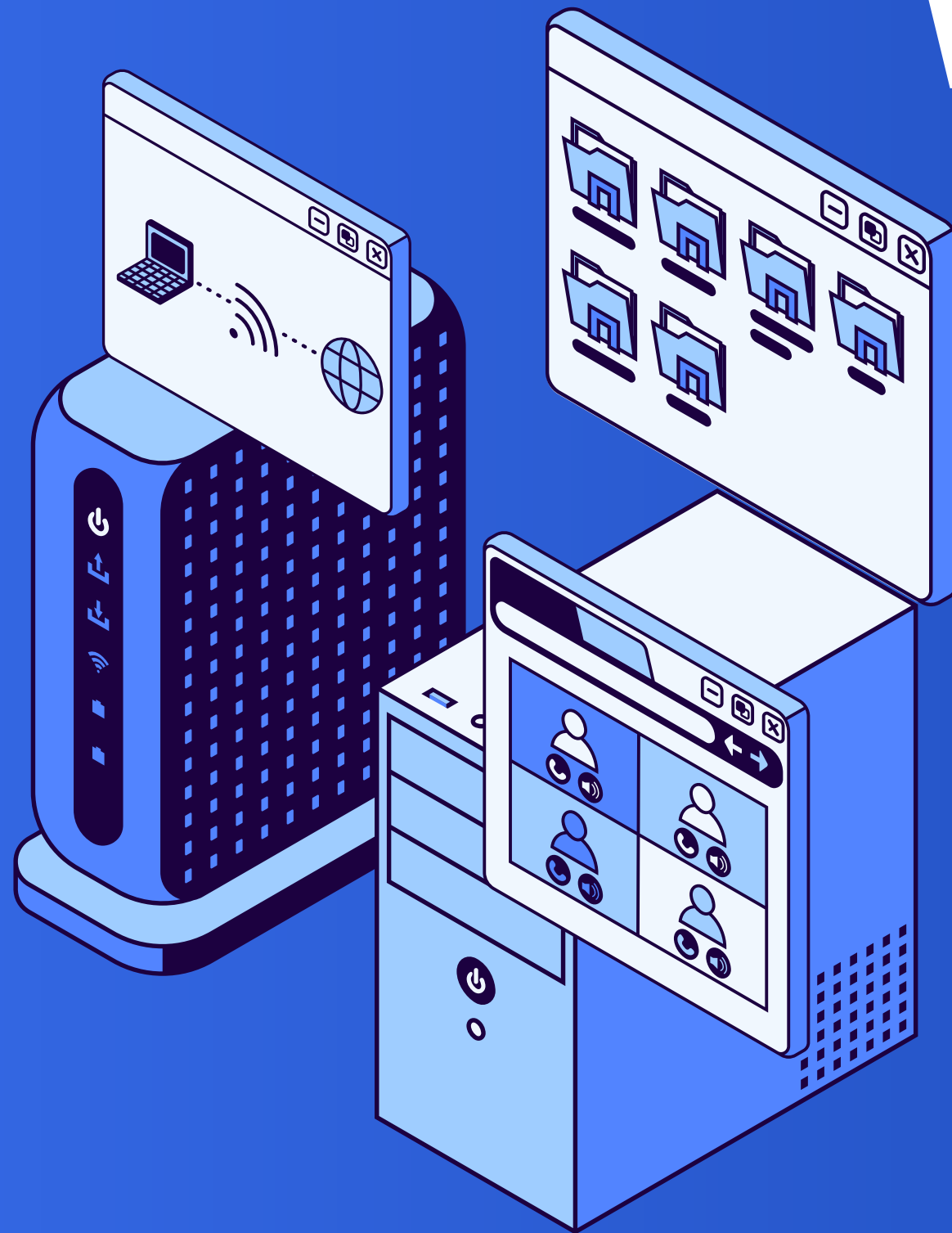
Components Required



- Arduino Uno
- GSM SIM800 / SIM800C Module
- PIR Motion Sensor
- SIM Card
- Jumper Wires
- Power Supply (5V)
- Mobile Phone



Working Principle



The PIR motion sensor detects human movement.
The PIR sensor sends a signal to the Arduino Uno.
Arduino processes the signal.
Arduino sends AT commands to the GSM SIM800 module.
The GSM module:

- Sends an SMS alert
- Makes a phone call to the stored mobile number.

The user receives the notification and can take action.



PIR Motion Sensor

- The PIR sensor detects infrared radiation emitted by humans or animals. When motion is detected, the sensor outputs a HIGH signal to the Arduino.

GSM SIM800 Module

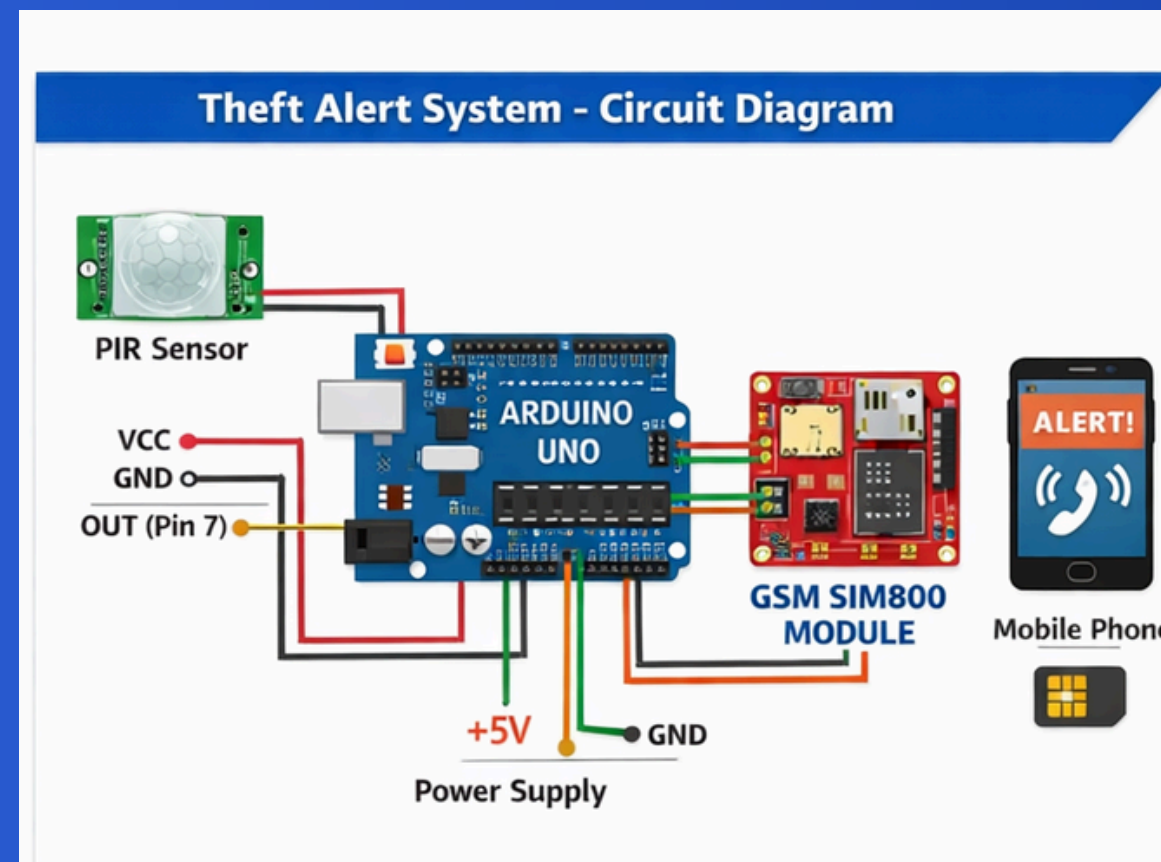
- The GSM SIM800 module is used for wireless communication. It allows the system to send SMS messages and make phone calls.

Arduino UNO

- Arduino Uno acts as the brain of the system. It reads the signal from the PIR sensor and controls the GSM module.

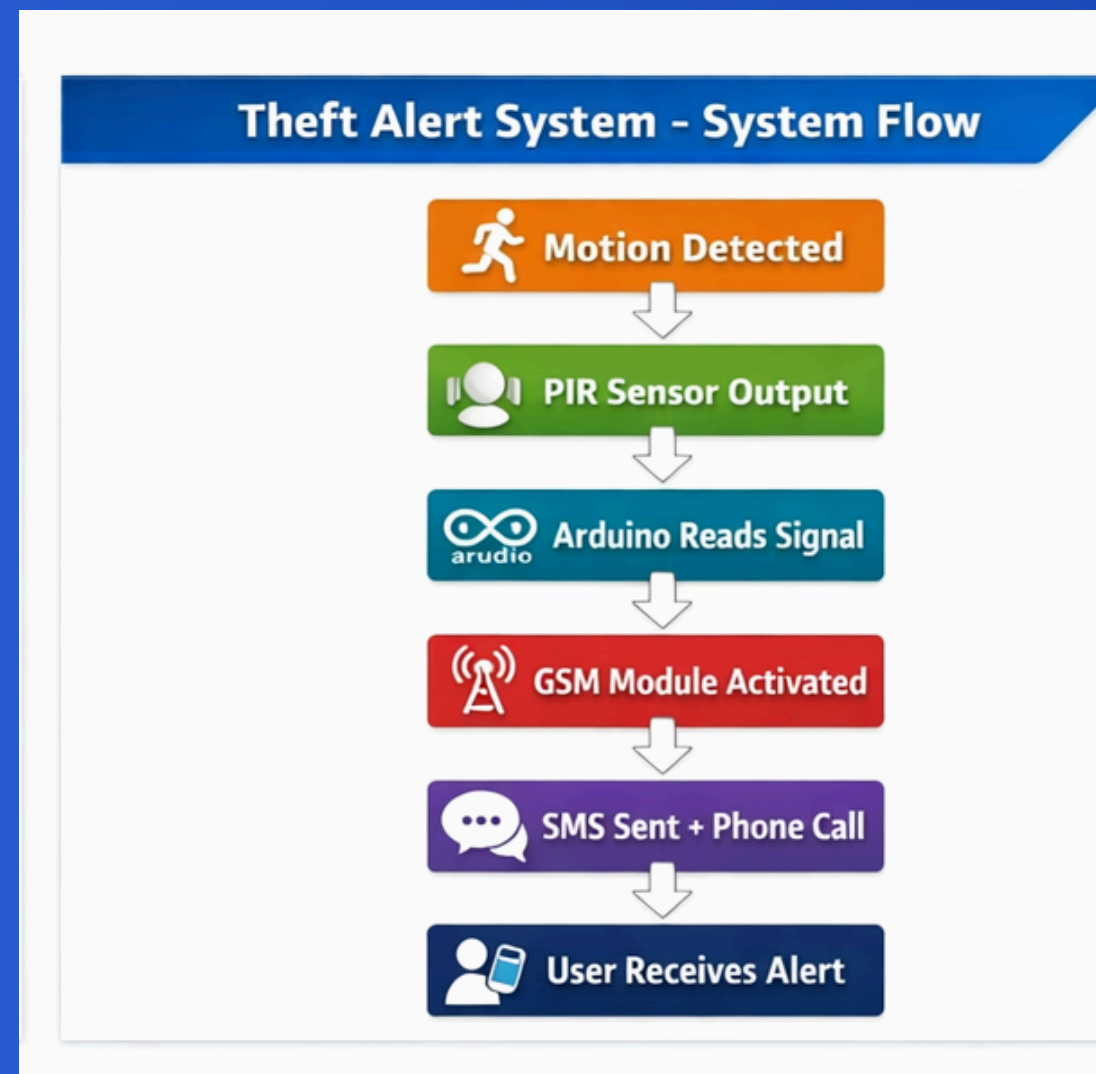


Circuit Diagram





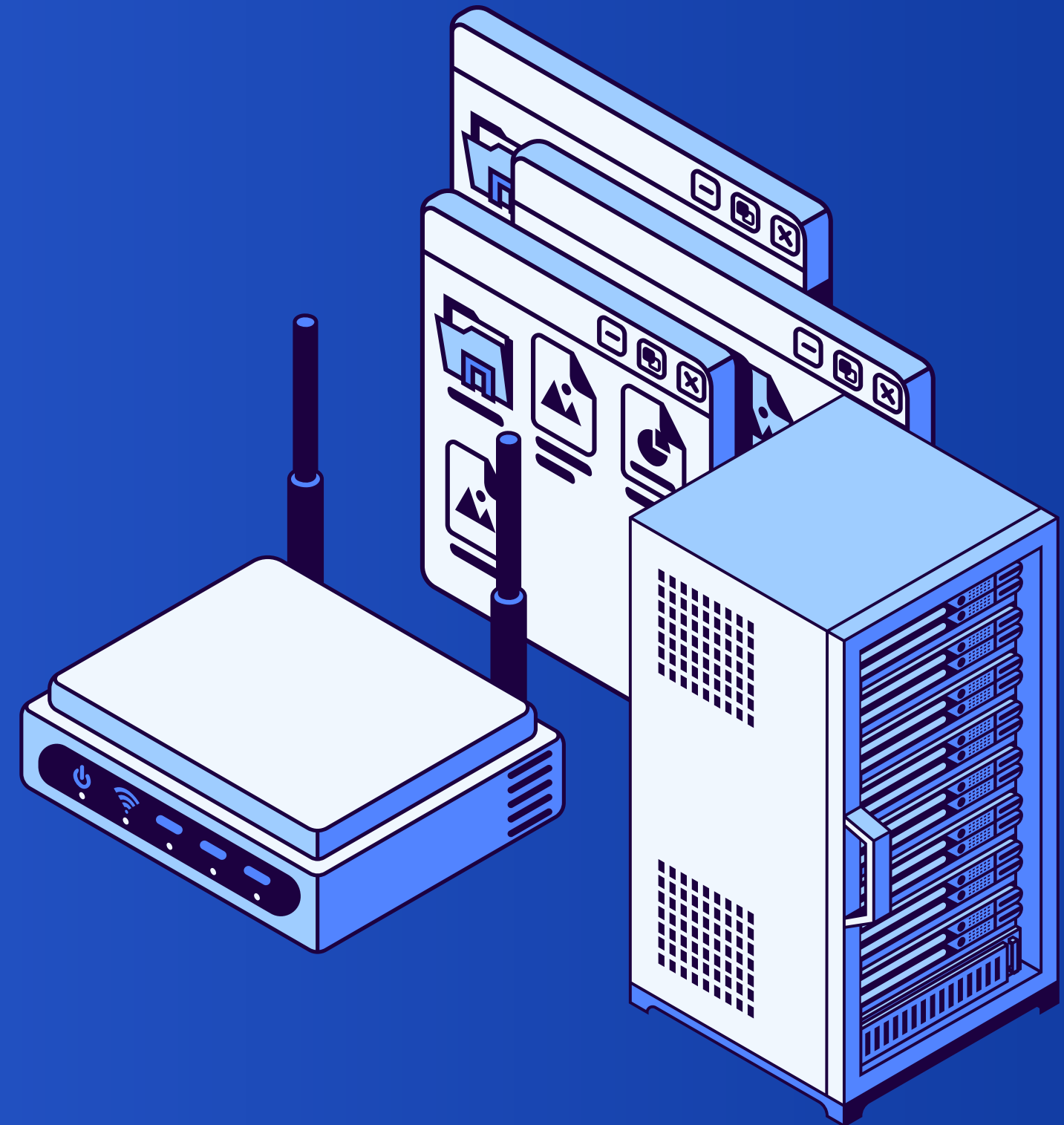
System Flow

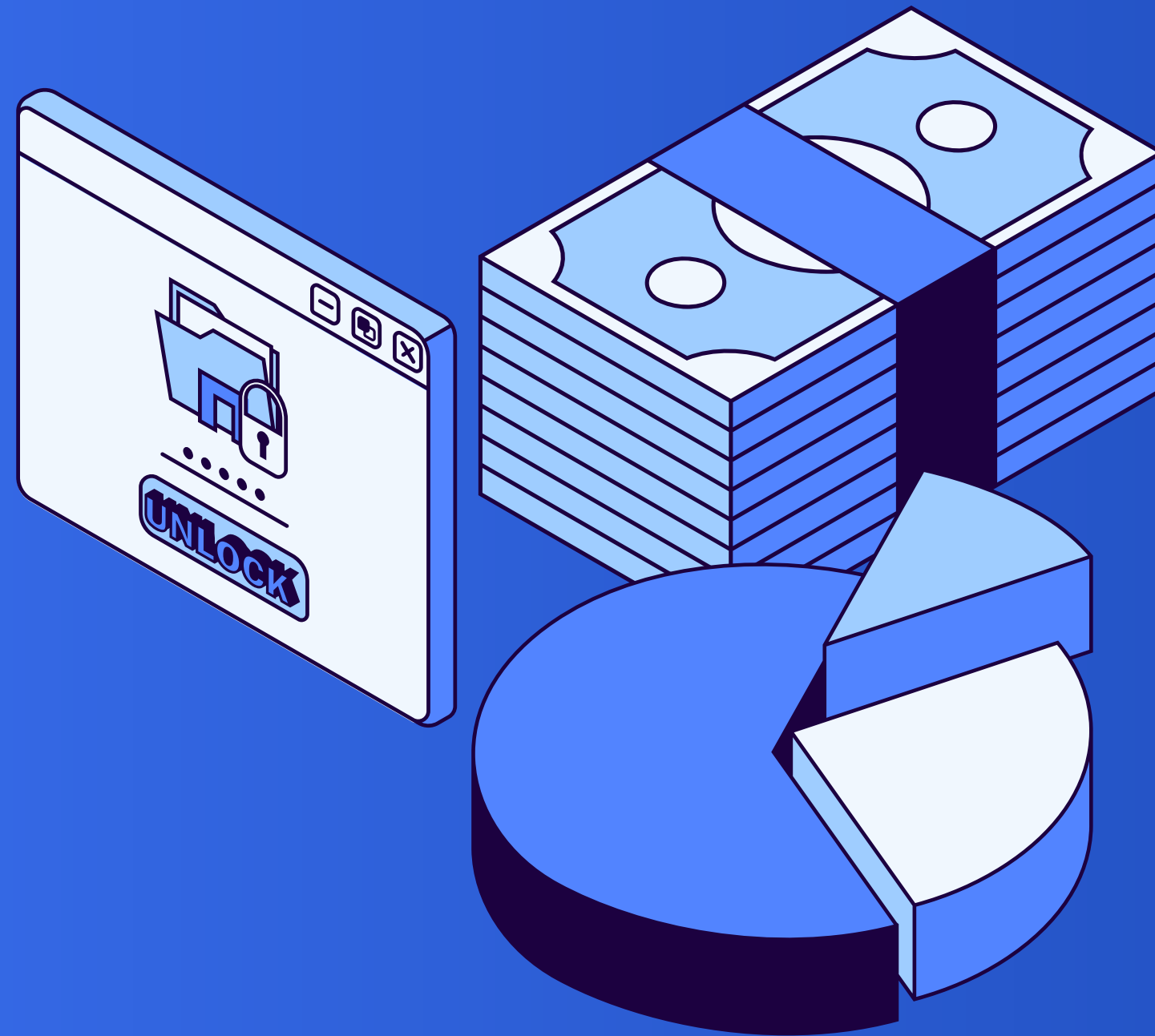




Advantages

- Low-cost security system
- Easy to install and maintain
- Instant alert through mobile phone
- Works without internet connection
- Reliable GSM communication





Limitations

- Depends on GSM network signal
- Limited detection range of PIR sensor
- False detection possible in some conditions
- Requires continuous power supply



Future Enhancements

The system can be upgraded by adding:

- ESP32-CAM for live video streaming
- IoT mobile application control
- Cloud data storage
- Alarm siren system
- Face recognition using AI



Applications

- Home security system
- Office security
- Bank locker monitoring
- Vehicle anti-theft system
- Warehouse security





Conclusion

- The Theft Alert System using GSM SIM800 module is an effective security solution. It detects motion and immediately alerts the user through SMS and phone calls. This system can be further improved by adding a camera, IoT features, or mobile app control.



Thank You